



Aprendizagem 2025
**Labs 6-7: Perceptron and
Gradient Descent**

Practical exercises

I. Perceptron

1. Considering the following linearly separable training data

	y_1	y_2	y_3	output
x_1	0	0	0	-1
x_2	0	2	1	+1
x_3	1	1	1	+1
x_4	1	-1	0	-1

Given the perceptron learning algorithm with a learning rate of 1 for simplicity, sign activation, and all weights initialized to one (including the bias).

- Considering y_1 and y_2 , apply the algorithm until convergence.
Draw the separation hyperplane.
- Considering all input variables, apply one epoch of the algorithm.
Do weights change for an additional epoch?
- Identify the perceptron output for $x_{new} = [001]^T$
- What happens if we replace the sign function by the step function?

$$\Theta(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

Specifically, how would you change the learning rate to get the same results?

- Show graphically, instantiating the parameters, that a perceptron:
 - can learn the following logical functions: NOT, AND and OR
 - cannot learn the logical XOR function for two inputs

II.

Gradient descent learning

Considering the following training data

	y1	y2	output
x_1	1	1	1
x_2	2	1	1
x_3	1	3	0
x_4	3	3	0

3. Let us consider the following activation

$$\hat{z} = output(\mathbf{x}, \mathbf{w}) = \frac{1}{1 + \exp(-2\mathbf{w} \cdot \mathbf{x})}$$

and half sum of squared errors as the loss function

$$E(\mathbf{w}) = \frac{1}{2} \sum_{k=1}^N (z_k - \hat{z}_k)^2, \text{ where } \hat{z}_k = output(\mathbf{x}_k, \mathbf{w})$$

- Determine the gradient descent learning rule for this unit.
- Compute the first gradient descent update assuming an initialization of all ones
- Compute the first stochastic gradient descent update assuming an initialization of all ones.

4. Let us consider the following function:

$$output(x, w) = \frac{1}{1 + \exp(-w \cdot x)}$$

and the cross-entropy loss function

$$E(\mathbf{w}) = -\log(\mathbf{z} | \mathbf{w}) = - \sum_{k=1}^N (z_k \log(\hat{z}_k) + (1 - z_k) \log(1 - \hat{z}_k))$$

- Determine the gradient descent learning rule for this unit
- Compute the first gradient descent update assuming an initialization of all ones
- Compute the first stochastic gradient descent update assuming an initialization of all ones

5. Let us consider the following function:

$$output(x, w) = \exp((w \bullet x)^2)$$

and half sum of squared errors as the loss function

a) Determine the gradient descent learning rule for this unit.

b) Compute the stochastic gradient descent update for input

$x_{new} = [11]^T$, $z_{new} = 0$ initialized with $w = [010]^T$ and learning rate $\eta=2$

6. Consider the sum squared and cross-entropy losses. Any stands out?

What changes when one changes the loss function?